

CSE4007 Artificial Intelligence - Assignment #1

Due: April 6th, 2026 11:59 pm

The submission should include a code (both link to the colab *with full access permission* and .py format) and a report that has answers to the questions and results. Use **PyTorch**. Also, minimize the use of numpy. There will be mark deductions if numpy is used when PyTorch is preferable or should be used. Marks will be deducted if the submission does not include the requested files. **DO NOT** use other libraries, such as scikit-learn/sklearn, to use a model you are supposed to implement. **Using sklearn or any other third library or already built-in functions** that you are asked to implement **will result in 0 mark**. Any form of **plagiarism** (e.g., using ChatGPT, copying works from others or the Internet, etc.) will result in **an immediate zero** for this assignment. Implement **all the TODO sections** indicated in the assignment code. Just a small tip: it's always a good idea to verify the implementation for each module separately. Using visualizations is one of ways to check the correctness of the implementation. *Use this link to download the assignment code.*

Problem #1: Preparation of VOC Datasets and Visualization

- a) Prepare the VOC datasets (training set and test set). Implement and use **Augmenter** and **Normalizer**.
- b) Implement the **VOCDataset** class. To do this, examine the VOC dataset directly and handle it appropriately.
- c) Visualize images using the prepared **VOCDataset** instance.
- d) We use a **collater()** function to create the VOC Dataloader. Explain why the **collater()** function is necessary.
- e) Complete the **collater()** function and initialize the Dataloaders using it.

Problem #2: Building Models and Anchors

- a) Implement the architecture of the Feature Pyramid Network. Note that the Feature Pyramid Network uses ResNet101 as its backbone in this assignment.
- b) Implement the forward pass of the Feature Pyramid Network.
- c) Implement the architecture and forward pass of the Regression Model (box subnet).
- d) Implement the architecture and forward pass of the Classification Model (class subnet).
- e) Implement the `BBoxTransform`, `Anchors` classes and `generate_anchors()`, `shift()` functions.

Problem #3: Training Models

- a) Train RetinaNet using BCE loss. Complete the training loop.
- b) Visualize the test set using the trained RetinaNet and compare it with its ground truth.
- c) Train RetinaNet using focal loss. Complete the training loop.
- d) Visualize the test set using the trained RetinaNet and compare it with its ground truth.
- e) Describe the differences in the inference results of the models when using BCE loss versus focal loss.